Exercises

HUJI 67594 · updated weekly

How to use. Each week has its questions first and its solutions on a separate page, so you can work without seeing the answers. Difficulty: ● ○ ○ direct use of a formula, ● ● ○ several steps, ● ● ● exam level. Sources: [KR 6.P11] = Kurose & Ross, 8th ed., chapter 6, problem 11 (reworded); [R2], [T1] = a variant of a recitation or tutorial question with new numbers; [own] = written for this sheet. The formulas are in `formula_sheet.pdf`; the explanations are in `weekNN.pdf`.

Contents. Week 01: delay, switching, probability (14 questions). Week 02: layering, ALOHA, CSMA/CD, WiFi (18 questions).

WEEK 01 — Lecture 1: Introduction · Tutorial 1: Probability

Delay and switching

Q1.1 ● ○ ○ [KR 1.P6]

Hosts A and B are m metres apart, joined by one link of rate R bit/s with propagation speed s. A sends one packet of L bits starting at $t = 0$. (a) Write d_{prop}, d_{trans} and the end-to-end delay (no processing or queuing). (b) At $t = d_{\text{trans}}$, where is the last bit? (c) If $d_{\text{prop}} > d_{\text{trans}}$, where is the first bit at $t = d_{\text{trans}}$? (d) With $s = 2.5 \times 10^8$ m/s, $L = 1500$ bytes and $R = 10$ Mbps, for which m is $d_{\text{prop}} = d_{\text{trans}}$?

Q1.2 ● ● ○ [KR 1.P10]

A 1500-byte packet crosses three links joined by two switches. Every link has $R = 2.5$ Mbps and $s = 2.5 \times 10^8$ m/s; the links are 5000, 4000 and 1000 km long. Each switch adds a processing delay of 3 ms. There is no queuing. Find the end-to-end delay.

Q1.3 ● ○ ○ [KR 1.P12]

A packet arrives at a router. On its outgoing link one packet is half transmitted and four more are waiting. All packets are 1500 bytes and the link is 2.5 Mbps. (a) What is the queuing delay of the new packet? (b) Give the general formula when x bits of the current packet are already sent and n packets wait.

Q1.4 ● ● ○ [KR 1.P31]

A message of 10^6 bits goes from a source through two store-and-forward switches to a destination (three links, 5 Mbps each). Ignore propagation, queuing and processing. (a) How long does it take without segmentation? (b) Now it is cut into 100 packets of 10 000 bits. When does the first packet reach the first switch? When is the second packet fully received there? (c) How long does the whole message take now? Why is it faster?

Q1.5 ● ○ ○ [own]

A 1 000 000-bit file is sent over a circuit-switched path. Every link is a 2 Mbps TDM link with 20 slots, and setting up the circuit takes 300 ms. How long until the whole file has arrived (ignore propagation)? Does it depend on the number of links?

Q1.6 ● ○ ○ [own]

An application has peak rate 1 Mbps and average rate 50 kbps. (a) What is its utilization under circuit switching? (b) How many such applications fit on a 10 Mbps link with circuit switching? (c) How many would fit if each needed only its average rate?

Q1.7 ● ○ ○ [own]

Packets of 1000 bits arrive at a 1 Mbps link at $a = 900$ packets/s on average. (a) What is the traffic intensity? (b) Up to which arrival rate is the queue stable?

Q1.8 ●●○ [KR 1.P8]

Users share a 10 Mbps link. Each needs 200 kbps when active and is active 10% of the time, independently. (a) How many users does circuit switching support? (b) With packet switching and 120 users, what is the probability that exactly n users are active? (c) What is the probability that 51 or more are active? What does that mean for packet switching?

Q1.9 ●●○ [KR 1.P9]

Same as Q1.8 but with a 1 Gbps link and users of 100 kbps, each active with probability $p = 0.1$. (a) What is N, the number of users under circuit switching? (b) With packet switching and M users, write the probability that more than N are active.

Probability

Q1.10 ●●○ [T1 / lecture Example II]

Good packets survive t seconds with probability e^{-t}, bad ones with probability e^{-1000t}. A packet is good with probability 0.2. (a) What is the probability that a packet survives 2 ms? (b) It survived 2 ms: what is the probability that it is good?

Q1.11 ●○○ [KR 1.P22]

A path has 5 links, each losing a packet independently with probability 0.05. The server resends a lost packet end to end. (a) Probability that one transmission arrives? (b) Expected number of *re*transmissions?

Q1.12 ●●● [T1 / lecture Example III]

A chain of $n = 4$ links, each failing independently with probability $p = 0.2$. (a) With end-to-end retransmission, what is the expected number of attempts by S, and the expected number of *link* transmissions? (b) With hop-by-hop retransmission, what is the expected number of link transmissions? Which scheme is cheaper?

Q1.13 ●●○ [own]

Packets arrive as a Poisson process with rate $\lambda = 3$ per second. (a) Probability of no packet in 1 s? (b) Probability of exactly 2 packets in 0.5 s? (c) You have already waited 2 s without a packet. What is the expected remaining wait? Why?

Q1.14 ●○○ [T1]

In each interval of length T the number of packets sent is Pois(gT). (a) What is the probability of exactly one packet? (b) For which gT is it largest, and what is the largest value?

WEEK 01 · SOLUTIONS

Q1.1

(a) $d_{\text{prop}} = m/s$, $d_{\text{trans}} = L/R$, end-to-end $= L/R + m/s$. (b) It is just leaving A. (c) The first bit is still on the link, $s \cdot d_{\text{trans}}$ metres from A; it has not reached B. (d) $m = s \cdot L/R = 2.5 \times 10^8 \cdot 12\,000/10^7 = 2.5 \times 10^8 \cdot 1.2 \times 10^{-3} = 300\,\text{km}$.

Q1.2

Each link: $L/R = 12\,000/2.5 \times 10^6 = 4.8\,\text{ms}$; three links: 14.4 ms (store-and-forward transmits on every link). Propagation: $20 + 16 + 4 = 40\,\text{ms}$. Processing: $2 \times 3 = 6\,\text{ms}$. Total $= 14.4 + 40 + 6 =$ **60.4 ms**.

Q1.3

(a) Half a packet plus four packets: $(0.5 + 4) \times 4.8\,\text{ms} =$ **21.6 ms**. (b) $\dfrac{nL + (L - x)}{R}$.

Q1.4

(a) Each link takes $10^6/(5 \times 10^6) = 0.2\,\text{s}$; store-and-forward over 3 links: **0.6 s**. (b) One packet per link: $10^4/(5 \times 10^6) = 2\,\text{ms}$. The first packet reaches switch 1 at 2 ms; the second is fully received there at 4 ms. (c) $(N + K - 1)\,L/R = (3 + 100 - 1) \times 2\,\text{ms} =$ **204 ms**, about 3 times faster: the links work in parallel (a pipeline) instead of waiting for the whole message.

Q1.5

Each circuit gets $2\,\text{Mbps}/20 = 100\,\text{kbps}$, so transmission takes $10^6/10^5 = 10\,\text{s}$. Total $= 10 + 0.3 =$ **10.3 s**. It does not depend on the number of links: the bits stream through the reserved path without store-and-forward.

Q1.6

(a) $A/P = 50/1000 =$ **5%**. (b) Reserve the peak: $10/1 =$ **10**. (c) $10\,000/50 =$ **200**. Statistical multiplexing aims at something close to (c).

Q1.7

(a) $La/R = 1000 \cdot 900/10^6 =$ **0.9**. (b) It must stay below 1: $a < 10^6/1000 =$ **1000** packets/s. At 0.9 the queue is stable but the queuing delay is already large.

Q1.8

(a) $10\,\text{Mbps}/200\,\text{kbps} =$ **50** users. (b) $X \sim \text{Bin}(120, 0.1)$: $P(X = n) = \binom{120}{n} 0.1^n 0.9^{120-n}$. (c) $P(X \geq 51) = \sum_{n=51}^{120} \binom{120}{n} 0.1^n 0.9^{120-n} \approx 2 \times 10^{-20}$. The mean is 12 and the standard deviation $\sqrt{120 \cdot 0.1 \cdot 0.9} \approx 3.3$, so 51 is about 12 standard deviations away. Packet switching serves 120 users where circuit switching serves 50, and the link is practically never overloaded.

Q1.9

(a) $N = 10^9/10^5 =$ **10 000**. (b) $P(X > N) = \sum_{k=N+1}^{M} \binom{M}{k} p^k (1-p)^{M-k}$, with $X \sim \text{Bin}(M, p)$.

Q1.10

Let $A =$ "survives 2 ms", $B =$ "good". (a) Total probability: $P(A) = 0.2e^{-0.002} + 0.8e^{-2} = 0.1996 + 0.1083 =$ **0.308**. (b) Bayes: $P(B \mid A) = 0.1996/0.308 =$ **0.648**. Surviving raised the chance of "good" from 0.2 to 0.65.

Q1.11

(a) All 5 links must work (independent, so multiply): $0.95^5 =$ **0.774**. (b) Attempts $\sim$ Geo(0.774), mean $1/0.774 = 1.292$, so on average **0.29** retransmissions (= attempts -1).

Q1.12

(a) An attempt succeeds with probability $0.8^4 = 0.4096$, so $\mathbb{E}[\text{attempts}] = 1/0.4096 =$ **2.44**. An attempt uses link k only if links $1..k-1$ worked: $\mathbb{E}[\text{links per attempt}] = 1 + 0.8 + 0.64 + 0.512 = 2.952$. Total $2.44 \times 2.952 \approx$ **7.2** link transmissions. (b) Each hop needs Geo(0.8) sends, mean 1.25; four hops: **5**. Hop-by-hop is cheaper because a loss only repeats one link, not the whole path.

Q1.13

(a) Pois(3): $e^{-3} = \mathbf{0.050}$. (b) Pois(1.5): $e^{-1.5}\,1.5^2/2 = \mathbf{0.251}$. (c) $\mathbf{1/3\,s}$. Inter-arrival times are Exp(3), which is memoryless: the 2 s already waited change nothing.

Q1.14

(a) $P(X = 1) = gT\,e^{-gT}$. (b) The derivative $(1 - gT)e^{-gT}$ is zero at $gT = 1$, where the value is $1/e \approx \mathbf{0.368}$. This is slotted ALOHA's best goodput (week 2).

WEEK 02 Lecture 2: Layering & Broadcast · Recitation 2: ALOHA

Layering

Q2.1 ●○○ [own]

An application sends 200 bytes over TCP/IP/Ethernet. (a) What fraction of the headers-plus-data is useful data? (b) What if you also count Ethernet's 4-byte CRC and 8-byte preamble?

Q2.2 ●○○ [R2]

Give the OSI layer of: HTTP, TLS encryption, TCP acknowledgements, IP routing, CSMA/CD, the optical fibre. Which of them are not a separate layer in the Internet's 5-layer stack?

Channel partitioning and ALOHA

Q2.3 ●○○ [own]

A 40 Mbps channel is shared by TDMA among 8 stations; only 3 have data. (a) Rate of each busy station? (b) How much capacity is wasted? (c) Which wish-list property fails?

Q2.4 ●●○ [KR 6.P8, P9]

(a) For slotted ALOHA with N nodes, find the p that maximizes $Np(1-p)^{N-1}$ and the limit of the maximum as $N \to \infty$. (b) Do the same for pure ALOHA, $Np(1-p)^{2(N-1)}$.

Q2.5 ●●○ [KR 6.P11]

Four nodes A, B, C, D use slotted ALOHA; each always has frames and sends in each slot with probability p. Slots are numbered $1, 2, \ldots$ (a) Probability that A succeeds for the first time in slot 4? (b) Probability that some node succeeds in slot 5? (c) Probability that the first success (by anyone) is in slot 4? (d) Efficiency of the system? Give all answers as formulas and as numbers for $p = 0.25$.

Q2.6 ●●● [KR 6.P10]

Two nodes A and B use slotted ALOHA with probabilities p_A and p_B. (a) Give A's average throughput (fraction of slots A wins) and the total efficiency. (b) If $p_A = 2p_B$, is A's throughput twice B's? If not, which p_A makes it exactly twice? (c) With N nodes, A using $2p$ and all others p: the throughput of A and of any other node?

Q2.7 ●●○ [R2]

A slotted ALOHA channel has $B = 2$ Mbps, frames of $L = 2000$ bits, and $n = 8$ nodes, each sending with $p = 0.1$. (a) Slot length? (b) Goodput? (c) Throughput (fraction of slots with any transmission)? (d) Useful bit rate? (e) Average time between successful frames?

Q2.8 ●●○ [R2 / own]

A channel carries on average $G = 0.4$ attempts per slot (Poisson model). (a) Slotted ALOHA: fraction of slots that are successful, empty, and collisions? (b) Pure ALOHA goodput? (c) Should the load be increased or decreased to maximize each?

Q2.9 ●●● [R2 Q1 variant]

Hosts C and D are both attached to two networks. The upper one runs slotted ALOHA and has 3 other nodes; the lower one runs pure ALOHA and has 2 other nodes. D never sends. Every other node sends in each of its slots with probability q. When C has a message for D it picks the upper network with probability p and then sends with probability q. (a) C has one attempt: probability that the message arrives? Numbers for $p = 0.6$, $q = 0.1$. (b) C now tries at the start of every slot (choosing a network each time). Goodput of the upper network? Same numbers.

Q2.10 ●●● [R2 Q2 variant]

One channel carries n slotted-ALOHA nodes (slots of length T, aligned to a common grid) and m pure-ALOHA nodes (frames of length T, own clocks, not aligned with the grid). Each node sends with probability p at the start of each of its slots. (a) Probability that a given slotted node succeeds in a slot? (b) That a given pure node succeeds? (c) Numbers for $n = m = 5$, $p = 0.1$. Who does better, and why?

CSMA/CD, Ethernet, WiFi

Q2.11 ●○○ [own]

A CSMA/CD LAN runs at 1 Gbps over at most 200 m of cable ($s = 2 \times 10^8$ m/s). What is the minimum frame size? Compare with Ethernet's 64 bytes.

Q2.12 ●○○ [KR 6.R6]

After the 5th collision, what is the probability that a node picks $K = 4$? How long is that delay on 10 Mbps Ethernet?

Q2.13 ●○○ [KR 6.P17]

For $K = 100$, how long does an adapter wait on a 100 Mbps channel? On a 1 Gbps channel?

Q2.14 ●●○ [KR 6.P18]

A and B are on a 10 Mbps channel, 325 bit times apart. A starts sending a minimum-size frame ($512 + 64 = 576$ bit times including the preamble) at $t = 0$; B starts before A's signal reaches it. Can A finish its frame before it hears B? What does that mean?

Q2.15 ●●○ [own]

Two NICs have just collided for the 2nd time. (a) From which set is K chosen? (b) Probability that they collide again? (c) Probability they collide on each of the next three attempts (3rd, 4th and 5th)?

Q2.16 ●●○ [lecture]

On a CSMA/CD LAN, PROP $= 2\,\mu$s and TRANSP $= 50\,\mu$s. (a) Efficiency in the simple model ($A = 1.5$) and the precise model? (b) The simple model with the large-N value $\alpha = 1/e$ instead of 0.4?

Q2.17 ●●● [KR 6.P20]

A CSMA/CD-like protocol: time is slotted with short slots; a frame lasts k slots. When the channel is free, each of N nodes transmits in a slot with probability p. If exactly one does, it holds the channel for its whole frame (k slots); otherwise the slot is wasted. (a) Efficiency for fixed N, p? (b) The best p? (c) The efficiency as $N \to \infty$? (d) What happens as $k \to \infty$?

Q2.18 ●○○ [own]

In 802.11, RTS and CTS take 25 μs each and a data frame takes 500 μs. (a) What is the reservation overhead per frame? (b) How much time does a collision waste with and without RTS/CTS? (c) When is RTS/CTS not worth it?

WEEK 02 · SOLUTIONS

Q2.1

(a) $200/(200+54) = \mathbf{78.7\%}$. (b) $200/(200+54+4+8) = 200/266 = \mathbf{75.2\%}$.

Q2.2

HTTP: 7 (application). TLS: 6 (presentation). TCP ACKs: 4 (transport). IP routing: 3 (network). CSMA/CD: 2 (data link, MAC sub-layer). Fibre: 1 (physical). In the Internet stack, session (5) and presentation (6) are not separate layers: the application provides them, so TLS counts as "application".

Q2.3

(a) $40/8 = \mathbf{5\,Mbps}$ each, even though more is free. (b) $40 - 3\cdot 5 = \mathbf{25\,Mbps}$ idle. (c) Property 2: $M = 3$ busy nodes should get $R/M \approx 13.3$ Mbps each. (Property 1 also fails: a lone node would get only $R/8$.)

Q2.4

(a) $\frac{d}{dp}Np(1-p)^{N-1} = N(1-p)^{N-2}[(1-p)-(N-1)p] = 0 \Rightarrow p^* = 1/N$. Maximum $(1-\frac{1}{N})^{N-1} \to 1/e \approx 0.37$. (b) $\frac{d}{dp}Np(1-p)^{2N-2} = N(1-p)^{2N-3}[(1-p)-(2N-2)p] = 0 \Rightarrow p^* = \frac{1}{2N-1}$. Maximum $\frac{N}{2N-1}(1-\frac{1}{2N-1})^{2N-2} \to \frac{1}{2}\cdot\frac{1}{e} = 1/(2e) \approx 0.18$.

Q2.5

A wins a given slot with probability $q = p(1-p)^3$ ($= 0.1055$ at $p = 0.25$), and slots are independent. (a) Fail 3 times, then win: $(1-q)^3 q = \mathbf{0.0755}$. (b) Exactly one of four: $4p(1-p)^3 = \mathbf{0.422}$. (c) Let $s = 4p(1-p)^3$. No success in slots 1–3, then success: $(1-s)^3 s = 0.578^3 \cdot 0.422 = \mathbf{0.0815}$. (d) Efficiency $= s = \mathbf{0.422}$ (here $p = 1/N$, so this is the best possible for 4 nodes).

Q2.6

(a) A wins when it sends and B does not: $p_A(1-p_B)$. Total efficiency $= p_A(1-p_B) + p_B(1-p_A)$. (b) No: $\dfrac{2p_B(1-p_B)}{p_B(1-2p_B)} = \dfrac{2(1-p_B)}{1-2p_B} > 2$, because A's extra sending also blocks B more often. Exactly twice needs $p_A(1-p_B) = 2p_B(1-p_A)$, i.e. $p_A = \dfrac{2p_B}{1+p_B}$. Check with $p_B = 0.1$: $p_A = 0.182$; $0.182 \cdot 0.9 = 0.164 = 2\cdot 0.1 \cdot 0.818$. (c) A: $2p(1-p)^{N-1}$. Another node: it sends, A is silent, and the other $N-2$ are silent: $p(1-2p)(1-p)^{N-2}$.

Q2.7

(a) $T = L/B = 2000/(2\times 10^6) = \mathbf{1\,ms}$. (b) $\eta = 8\cdot 0.1\cdot 0.9^7 = \mathbf{0.383}$. (c) $1 - 0.9^8 = \mathbf{0.570}$: the channel is busy 57% of the time but useful only 38%. (d) $0.383 \times 2\,\text{Mbps} = \mathbf{765\,kbps}$. (e) $T/\eta = 1/0.383 = \mathbf{2.6\,ms}$.

Q2.8

(a) Success $Ge^{-G} = \mathbf{0.268}$; empty $e^{-G} = \mathbf{0.670}$; collision $1 - 0.670 - 0.268 = \mathbf{0.062}$. (b) $Ge^{-2G} = 0.4e^{-0.8} = \mathbf{0.180}$. (c) Slotted: increase towards $G^* = 1$ (max 0.368). Pure: decrease slightly, to $G^* = 0.5$ (max 0.184); 0.4 is already near it.

Q2.9

(a) Total probability over the network chosen. Upper (slotted): C sends and the 3 others are silent in that slot: $q(1-q)^3$. Lower (pure): each of the 2 others must skip both of its overlapping slots: $q((1-q)^2)^2 = q(1-q)^4$.

$$P = q\left[p(1-q)^3 + (1-p)(1-q)^4\right] = 0.1\left[0.6\cdot 0.729 + 0.4\cdot 0.656\right] = \mathbf{0.070}.$$

(b) If C picks the upper network (prob. p) there are 4 active nodes; otherwise 3. A slot succeeds if exactly one sends:

$$\eta^{\text{upper}} = p\cdot 4q(1-q)^3 + (1-p)\cdot 3q(1-q)^2 = 0.6\cdot 0.292 + 0.4\cdot 0.243 = \mathbf{0.272}.$$

(Note the difference: (a) is about C, one node, no $\binom{n}{1}$; (b) is about the *network*, any node.)

Q2.10

(a) The slotted node sends (p); the other $n-1$ slotted nodes are silent in this slot ($(1-p)^{n-1}$). A pure frame of length T starting anywhere in $(t-T, t+T)$ overlaps the slot; each pure node has exactly 2 slot starts there: $(1-p)^{2m}$.

$$P_{\text{slotted}} = p(1-p)^{n-1}(1-p)^{2m}.$$

(b) The other pure nodes: $(1-p)^{2(m-1)}$, as in plain pure ALOHA. The pure frame is not aligned with the grid, so it always straddles **two** grid slots: all n slotted nodes must skip both, $(1-p)^{2n}$.

$$P_{\text{pure}} = p(1-p)^{2(m-1)}(1-p)^{2n}.$$

(c) Slotted: $0.1 \cdot 0.9^{14} = \mathbf{0.0229}$. Pure: $0.1 \cdot 0.9^{18} = \mathbf{0.0150}$. The slotted node does better: it is exposed to one slot of each slotted node, while the unaligned pure frame is exposed to two slots of everyone.

Q2.11

PROP $= 200/(2 \times 10^8) = 1\,\mu$s. Need TRANSP $\geq 2\,$PROP $= 2\,\mu$s, so $L \geq 2 \times 10^{-6} \cdot 10^9 = 2000$ bits $=$ **250 bytes**. Ethernet's 64 bytes would last only $0.512\,\mu$s at 1 Gbps: too short to detect collisions over 200 m. (Shared-medium gigabit Ethernet therefore had to extend short frames; in practice gigabit Ethernet is switched.)

Q2.12

After 5 collisions $K \in \{0, \ldots, 31\}$, uniform: $P(K=4) = \mathbf{1/32}$. Delay $= 4 \cdot 512 \cdot 0.1\,\mu\text{s} = \mathbf{204.8}\,\mu$**s**.

Q2.13

$100 \cdot 512 = 51\,200$ bit times. At 100 Mbps (10 ns/bit): **512** μ**s**. At 1 Gbps (1 ns/bit): **51.2** μ**s**.

Q2.14

Worst case: B starts just before A's first bit arrives, at $t \approx 325$. B's signal reaches A at $t \approx 325+325 = 650$ bit times. A finishes at $t = 576 < 650$, so **yes**: A finishes before it hears B, and wrongly believes its frame got through. The cable is too long for this frame size (TRANSP $<$ 2 PROP).

Q2.15

(a) $\{0,1,2,3\}$. (b) Same K: $4 \cdot (1/4)^2 = \mathbf{1/4}$. (c) The sets grow to 8, 16, 32 values: $\frac{1}{4} \cdot \frac{1}{8} \cdot \frac{1}{16} = \mathbf{1/512}$. (Strictly, the $K=0$ wait and "channel busy" rules can change this slightly; this is the usual exam answer.)

Q2.16

$a = 2/50 = 0.04$. (a) $1/(1+3a) = 1/1.12 = \mathbf{0.893}$; precise $1/(1+5a) = 1/1.2 = \mathbf{0.833}$. (b) $A = (1-\alpha)/\alpha = e - 1 \approx 1.72$, so $\eta = 1/(1+2Aa) = 1/(1+3.44 \cdot 0.04) = \mathbf{0.879}$.

Q2.17

(a) A contention slot is won with probability $\alpha = Np(1-p)^{N-1}$. The expected number of wasted slots before a win is $x = (1-\alpha)/\alpha$ (failures before a geometric success). Efficiency $= \dfrac{k}{k+x} = \dfrac{k}{k+(1-\alpha)/\alpha}$.

(b) Efficiency grows with α, so maximize α: $p^* = 1/N$. (c) $\alpha \to 1/e$, $x \to e-1$: efficiency $\to \dfrac{k}{k+e-1}$.

(d) $\dfrac{k}{k+e-1} \to \mathbf{1}$: long frames make the contention overhead negligible (the same as $a \to 0$ in the lecture).

Q2.18

(a) 50 μs per 500 μs frame: **10%**. (b) Without RTS/CTS a collision wastes a whole data frame, 500 μs; with it only an RTS, 25 μs. (c) When collisions are rare (light load) or frames are short: then the fixed 50 μs costs more than the collisions it prevents.